MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
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ICT 3143 Embedded Systems Lab
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MINI-PROJECT SYNOPSIS ON

CO Level Monitoring Using MQ-7

Submitted by

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DESCRIPTION

In the modern world, air pollution has become a significant environmental and health concern. Rapid industrialization, urbanization, and vehicular emissions have caused a sharp rise in the concentration of harmful gases, particularly **Carbon Monoxide (CO)**. Since CO is a **colourless, odourless, and highly toxic gas**, it poses a serious threat to human health even at low concentrations. Detecting and monitoring CO levels in real-time is therefore essential to ensure environmental safety and prevent health hazards.

The project titled *“CO Level Monitoring Using MQ-7”* focuses on developing an embedded system capable of **detecting and displaying CO concentration levels in real time**. The system uses an **MQ-7 gas sensor**, which is sensitive to carbon monoxide, and an **LPC1768 microcontroller** for data acquisition and processing. The analog output from the sensor is converted into a digital signal by the microcontroller’s inbuilt **ADC (Analog-to-Digital Converter)**. This digital data is then analysed and compared against predefined threshold limits to determine whether the air quality is within safe limits.

The processed data is displayed on a **16×2 LCD screen** that shows the current CO level (in parts per million, PPM) and air quality status. If the concentration exceeds the safe threshold, the system triggers a **buzzer alarm**, alerting users about poor air quality. This enables individuals to take immediate action, such as improving ventilation or moving to a safer environment.

The proposed system is **low-cost, compact, and energy-efficient**, making it suitable for both **domestic and industrial environments**. It can be installed in homes, offices, factories, or public spaces to continuously monitor CO levels. Furthermore, the system serves as a foundation for future advancements, it can be enhanced with **IoT connectivity** for cloud-based data storage, **mobile applications** for remote monitoring, and **machine learning algorithms** for predictive analysis of pollution trends.

In conclusion, this project provides a **reliable and affordable air quality monitoring solution** that not only enhances environmental awareness but also contributes to improving public health and safety. Through continuous CO monitoring and timely alerts, it supports the broader goal of building **smart and sustainable cities**.

METHODOLOGY

Connection Description:

- **MQ-7 Gas Sensor:** Provides an analogue output based on CO concentration; connected to the LPC1768's analogue input pin.
- **LPC1768 Microcontroller:** Converts analogue input to digital, processes data, and controls the LCD and buzzer.
- **LCD 16×2 Display:** Displays CO levels (in PPM) and air quality status.
- **Buzzer:** Sounds an alarm when pollutant concentration exceeds a threshold. P0.22 (use transistor driver if buzzer current exceeds GPIO capability).
- **5V DC Supply:** Powers all system components to ensure stable operation.

Ports Used:

1. **MQ-7 gas sensor**: Analog OUT → P0.23 (AD0.0). VCC → +5V, GND → GND.
2. LCD 16×2 (4-bit mode)
3. RS → P0.8.
4. EN → P0.9.
5. D4 → P0.4.
6. D5 → P0.5.
7. D6 → P0.6.
8. D7 → P0.7.
9. RW → GND (fixed).
10. **Buzzer / Alert:** Buzzer drive → P0.22 (use transistor driver if buzzer current exceeds GPIO capability)
11. **GPIO direction**: P0.4–P0.9 and P0.22 as outputs (FIODIR |= RS_CTRL | EN_CTRL | DT_CTRL | BUZZER_PIN).
12. **ADC initialization**: PINSEL1 for P0.23 → AD0.0 and enable ADC power/control bits in ADC registers.
13. **Power & common ground**: +5V supply → MCU Vcc and MQ-7 Vcc; all grounds common.

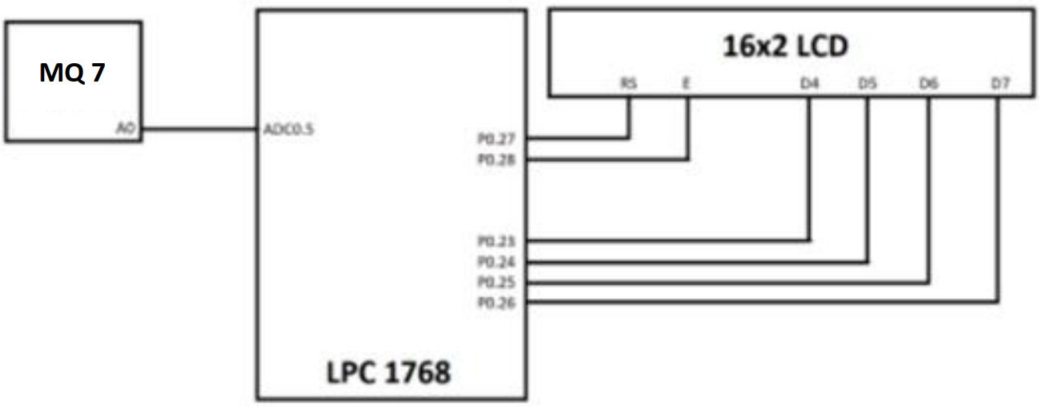

Fig . Circuit Diagram of the implementation

COMPONENTS REQUIRED

Hardware Requirements

1. MQ-7 Gas Sensor
2. LPC1768 Microcontroller
3. 16x2 LCD Display
4. Jumper Wires
5. Resistors
6. Transistor
7. External Power Supply
8. PCB (Printed Circuit Board) (optional)

Software Requirements

1. **Language**: C
2. **Application**: Keil Micro-Vision, Flash Magic